

Lab 2: Simple linear regression

Submit:

- Your notebook (within a zip file) to <http://deei-mooshak.ualg.pt/~jvo/ML/submissions/>
- Up to Oct 6, 2026

Distributed with these instructions is a Jupyter Notebook named **regression-intro.ipynb**, along with a dataset file named **demodataset.csv**. Download the notebook and the dataset file, inspect and execute all the code cells in the Jupyter Notebook.

Download the file **Lab2-regression.ipynb**, complete the notebook by answering the questions below, and submit it as your assignment in zip format.

Note: define functions when needed to prevent redundant code.

1.

- Compute θ_0^* and θ_1^* for the line of best fit:
 - using `scipy.stats.linregress()`
 - implementing the linear regression model below using **NumPy** functions:

$$\theta_0 = \bar{y} - \theta_1 \bar{x}$$

$$\theta_1 = \frac{SS_{xy}}{SS_{xx}} = \frac{\sum_{i=1}^m [(x_i - \bar{x})(y_i - \bar{y})]}{\sum_{i=1}^m (x_i - \bar{x})^2}$$

- check that i) and ii) produce the same results
- Print the dataset and superimpose the line of best fit.
 - Predict y for $x = 4.6$. What is the corresponding residual? Superimpose on the graphic obtained in b)
 - Interpret the parameters of line of best fit (θ_0, θ_1) in the context of the dataset. What does the slope tell you about the relationship between years of experience and salary? What does the intercept represent?
 - Use your final linear regression model to predict the salary for a person with 20 years of experience. Is this prediction an interpolation or an extrapolation? Briefly explain.

2.

- Express the cost function $J(\theta_0, \theta_1)$ in vector notation
 - Compute $J(\theta_0^*, \theta_1^*)$ for the given dataset
 - Show that point (average of X, average of Y) belongs to the line of best fit.

- b) For θ_0^* and θ_1^* , plot the residuals vs the independent variable x . Print the mean and variance of the residuals. Briefly comment on what you observe.
- c) For θ_0^* and θ_1^* , analytically prove that the sum of residuals is zero.
- d) Show that the derivative of the cost function, with respect to θ_j is:

$$\frac{\partial}{\partial \theta_j} J(\theta_0, \dots, \theta_n) = \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x_j^{(i)}) - y^{(i)}) x_j^{(i)} \quad , \quad j = 0, 1 \dots n$$

3.

- a) Express in vector notation the following gradient descent updating expressions:

$$\theta_j = \theta_j - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x_j^{(i)}) - y^{(i)}) x_j^{(i)} \quad , \quad j = 0, 1 \dots n$$

- b) Apply gradient descent and linear regression to the dataset. Find θ_0^* , θ_1^* and $J(\theta_0^*, \theta_1^*)$.
- c) What is the number of iterations and the value of the learning rate α , that approximate the θ vector and the cost J obtained in 3. b) to those found in 1. a) and 2. a)?
- d) Plot $J(\theta_0, \theta_1)$ as a function of the number of iterations. Briefly comment on what is observed.

Bibliography

[1] Jupyter notebook. <https://jupyter.org/>

[2] mathisfun. List of derivative rules:

<https://www.mathsisfun.com/calculus/derivatives-rules.html>